PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
DEPARTMENT OF COMPUTER APPLICATIONS
AY 2026-2027 MCA FIRST YEAR SEMESTER 1
26MX17 WEB APPLICATION DEVELOPMENT
Design Document

I. Team Detail

Team No.	Roll No.	Name of Student	Faculty Guide
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II. Title of the Application

UNITE

University Network for Interdisciplinary Teamwork and Engagement

III. Overview of the Project

UNITE (University Network for Interdisciplinary Teamwork and Engagement) is a college-restricted web application for students to form friendships and collaborate across academic programmes. Students create profiles containing About, Interests and custom Links, discover peers, and exchange friend requests. Students may create or join collaboration groups with a maximum of six members, with only one member from each academic programme. Group polls support collective decisions and meetups. Two or more existing groups can work together for events requiring larger teams, while each group retains its own membership limits. Students submit evidence of participation and achievement for faculty or administrator review. Only verified submissions qualify for group recognition and points-based leaderboards. UNITE encourages collaboration as both a friendly and an intellectual pursuit.

IV. Existing System

Platforms such as **LinkedIn** (https://www.linkedin.com/) support networking and profiles, while **Devpost** (https://devpost.com/) focuses on project and competition participation. Neither offers UNITE's proposed combination of programme-restricted groups, on-campus friendships, group polling and verified achievements in a single PSG-focused workflow.

V. Proposed System

- **1. Student Networking:** Student registration and login, profiles with About, Interests and customizable Links, peer discovery, and friend requests.
- **2. Interdisciplinary Groups:** Create or join groups of at most six students, with no duplicate academic programmes. Use polls to make group decisions, and permit separate groups to collaborate for larger events.
- **3. Verified Achievements:** Students submit participation details and evidence. Faculty or administrators approve or reject claims; approved achievements support group scores and leaderboards.

VI. Objectives of the Proposed Application

- Connect eligible students across academic programmes and interests.
- Enable friendships and simple, student-controlled profiles.
- Form valid interdisciplinary groups with a six-member maximum and one student per programme.
- Support poll-based group decisions and collaboration between existing groups.
- Provide a verifiable submission and review process for achievements.
- Recognise approved group achievements through transparent leaderboards.

VII. Scope / Use

- Student registration and profile management for eligible programmes; PhD/research programmes are excluded.
- Student discovery, friendship requests, and external profile links.
- Groups of up to six students, with one member per academic programme.
- Group polls for collective decisions and meetups; collaboration among independent groups.
- Student-submitted achievement evidence and authorised verification.
- Group leaderboard, basic notifications, and report/block options.

VIII. Technology to be Used

- **Front end:** HTML5, CSS3, vanilla JavaScript; Bootstrap if needed.
- **Back end:** Node.js and Express.js.
- **Database:** MySQL.
- **Security:** bcrypt password hashing, server-side sessions, input and authorization checks.

IX. Functional Requirements of the Application

- Students shall register, log in and edit About, Interests and custom Links.
- Students shall search peers and manage friend requests.
- The server shall enforce group size and unique-programme membership.
- Group members shall create polls and cast one vote per poll.
- Separate groups may collaborate on an event while retaining individual memberships.
- Students shall submit achievement evidence for faculty/admin review.
- Only approved achievements shall count toward the leaderboard.
- Students shall receive notifications and may report or block users.

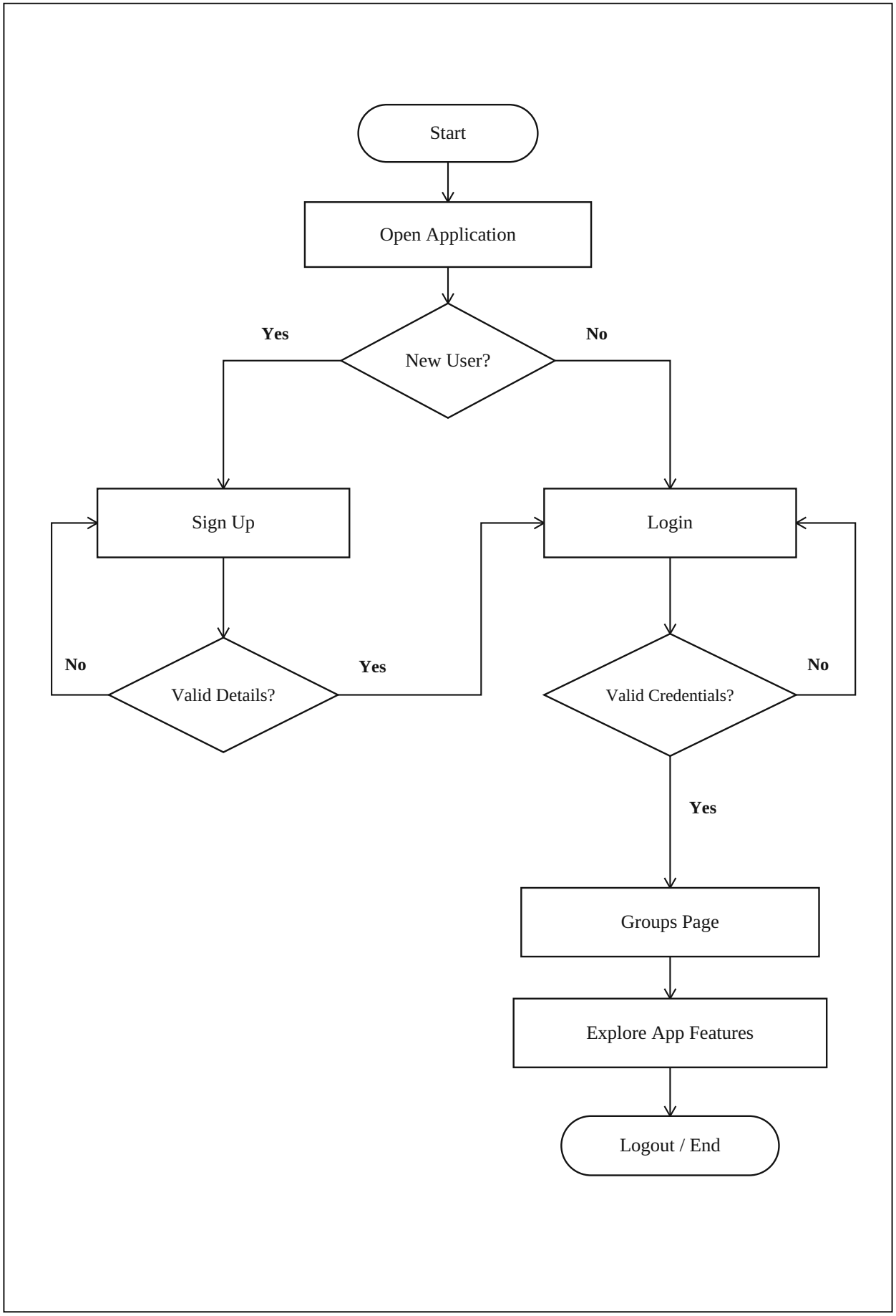
X. Non-Functional Requirements of the Application

- **Usability:** The interface shall be simple, responsive and usable on desktop and mobile browsers.
- **Performance:** Student search and group operations shall use efficient queries and page loading.
- **Reliability:** Membership and achievement updates shall avoid partial or inconsistent records.
- **Security:** Store password hashes, protect authenticated sessions, verify roles and validate uploads.
- **Integrity:** Check the six-member cap and programme uniqueness together in a server-side transaction.
- **Privacy:** Restrict access to personal information and submitted evidence as appropriate.
- **Maintainability:** Use modular code and technologies within the WAD syllabus.
- **Availability:** The deployed application shall remain accessible over supported browsers.

Design Constraints

Event category is determined by the **host institution**, not by venue: PSG-hosted = College (1x), Institute of National Importance-hosted = National (2x), and other-college-hosted = State (1.5x). Group credit for a jointly earned achievement is **not yet specified**; this document does not assume a point-splitting policy.

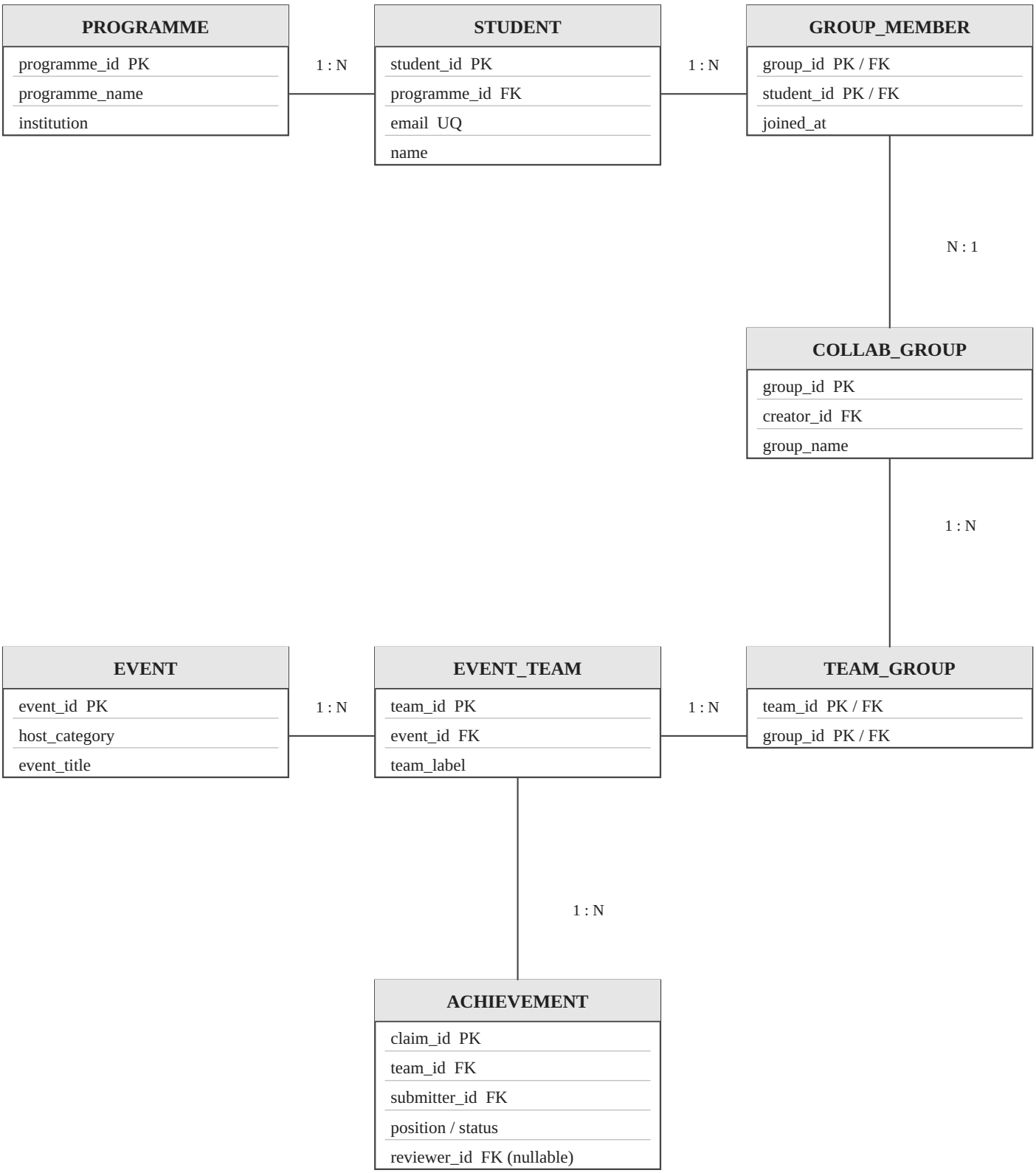
XI. Flow Diagram of the Application



XII. DATABASE SCHEMA OF THE APPLICATION

UNITE | Simplified core database schema

PK = Primary Key FK = Foreign Key UQ = Unique



Core rules: max 6 students per group; one student per programme.

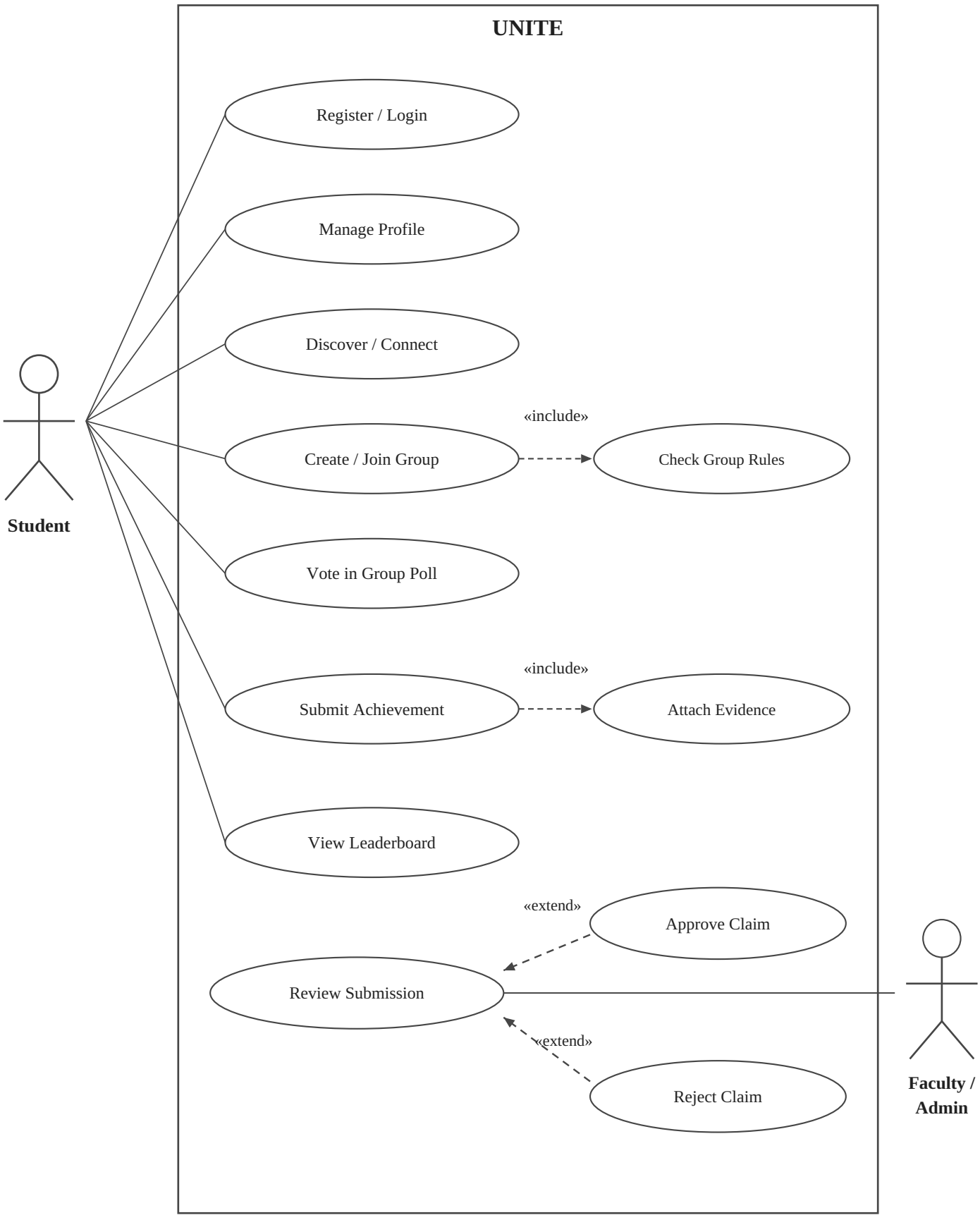
Joining checks run transactionally. Multiple groups may belong to one event team.

Supporting tables (profiles, friends, polls, votes, evidence) are intentionally not drawn.

This is a simplified diagram, not the complete SQL implementation schema.

XIII. USE CASE DIAGRAM OF THE APPLICATION

UNITE | UML use case diagram (include / extend within system boundary)



«include» = required action. «extend» = conditional outcome of review.

XIV. Module Identification

UNITE comprises the following main modules, designed to support student networking and interdisciplinary collaboration.

S.no	Module Name	Description
1	User Management	Student registration, secure login, logout and programme selection.
2	Profile and Discovery	Manage About, Interests and Links; search for students and view profiles.
3	Friendship Management	Send and respond to friend requests; manage connections and report or block users.
4	Group Management	Create, join and leave groups; check the six-member limit and programme uniqueness.
5	Poll and Collaboration	Conduct group polls, decide meetups and coordinate participation between groups.
6	Event Management	Record event details and associate one or more existing groups with an event.
7	Achievement Verification	Submit results and supporting evidence; allow faculty/admin approval or rejection.
8	Leaderboard and Alerts	Calculate scores for verified achievements, show rankings and notify students.

Module functions are implemented using the approved WAD technology stack.

XV. Timeline of Activities Planned / Completed

Activities Planned	Timeline
1. Requirement Analysis	02 Nov 2026
2. UI/UX Design	05 Nov 2026
3. Backend Development	13 Nov 2026
4. Integration & Testing	17 Nov 2026
5. Final Documentation	20 Nov 2026

Signature of the Students

Bala Subramanian B

Nareshwaran J

Venkatesh C R

Signature of the Guide

Dr. V. Umarani